

Assignment 12 – Selenium Page Object Model in Python Behave Framework

1. Module Name

Python BDD Restful Automations

2. Experiment Title

Scenario 3: Selenium Page Object Model in the Python Behave Framework

3. Problem Statement

Implement a maintainable Selenium automation framework using Page Object Model with Python and Behave so that page locators and UI actions are separated from business-readable BDD scenarios.

4. Objective

Create reusable page classes, centralize web element locators and actions, connect the page objects with Behave step definitions, and verify a successful SauceDemo login.

5. Tools, Software and Concepts Used

Python; Selenium WebDriver; Behave BDD; Google Chrome; Gherkin; Page Object Model; explicit waits; reusable page methods.

6. Source Code / Implementation Steps

Created LoginPage and InventoryPage classes containing locators and reusable methods. Created a Gherkin feature describing the login behavior. Implemented step definitions that instantiate and use the page objects. Added environment.py for WebDriver setup and teardown. Used explicit waits for reliable element interaction and verification.

7. Output / Labelled Screenshots

Screenshot 1 – Behave POM Execution: terminal output showing the successful scenario, steps passed and zero failures. The screenshot is stored in Source_Code/Screenshots.

8. Result

The Selenium Page Object Model framework was implemented successfully with Behave. The login scenario completed successfully using reusable LoginPage and InventoryPage methods.

9. Observation

POM separates page-specific implementation details from test behavior. Locators and UI actions can therefore be maintained centrally, while feature files remain readable and focused on business behavior.

10. Conclusion

The experiment successfully demonstrated the integration of Selenium Page Object Model with Python Behave BDD. The resulting structure provides a maintainable foundation for extending the automation framework.